

Lecture 7/Module 1

DC Circuit Analysis and Theorems

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Dr. Sudipta Kumar Sarkar

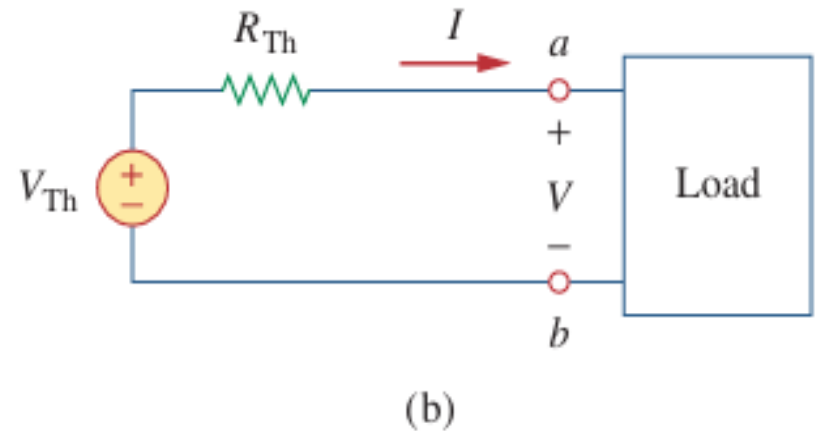
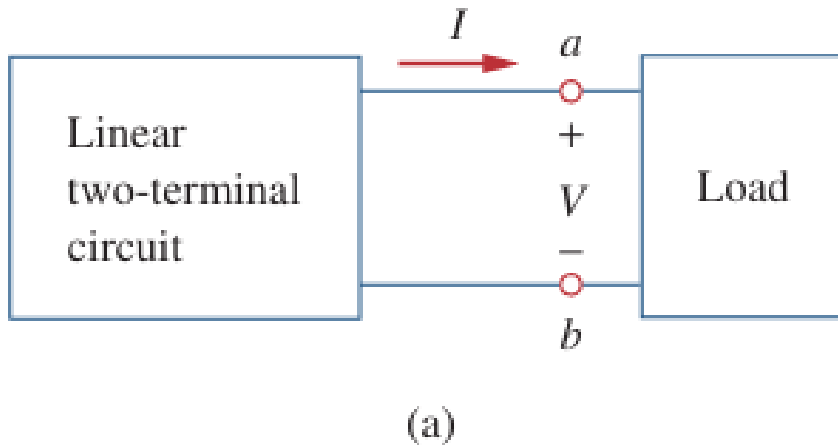
Assistant Professor (Research), SENSE



School of Electronics Engineering (SENSE)
VIT, Vellore, India

Thevenin's Theorem

Thevenin's theorem states that a linear two-terminal circuit can be **replaced by an equivalent circuit consisting of a voltage source V_{th} in series with a resistor R_{th}** , where V_{th} is the open-circuit voltage at the terminals and R_{th} is the input or equivalent resistance at the terminals when the independent sources are turned off.



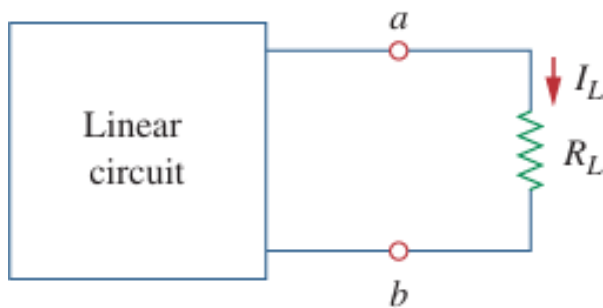
Two circuits are said to be equivalent if they have the same voltage-current relation at their terminals.

Thevenin's Theorem

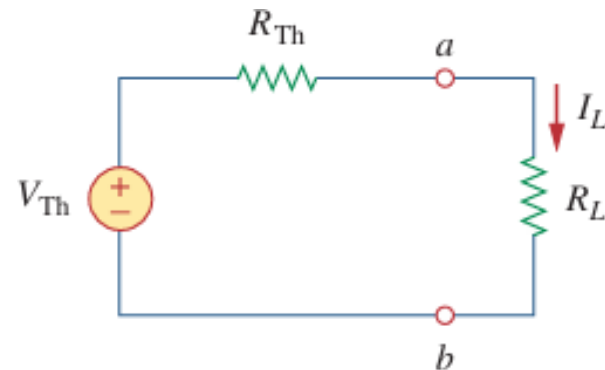
A large circuit may be replaced by a single independent voltage source and a single resistor.

A linear circuit with a variable load can be replaced by the Thevenin equivalent, exclusive of the load.

Consider a linear circuit terminated by a load R_L , the current through the load I_L and the voltage V_L across the load are easily determined once the Thevenin equivalent of the circuit at the load's terminals is obtained.



(a)



(b)

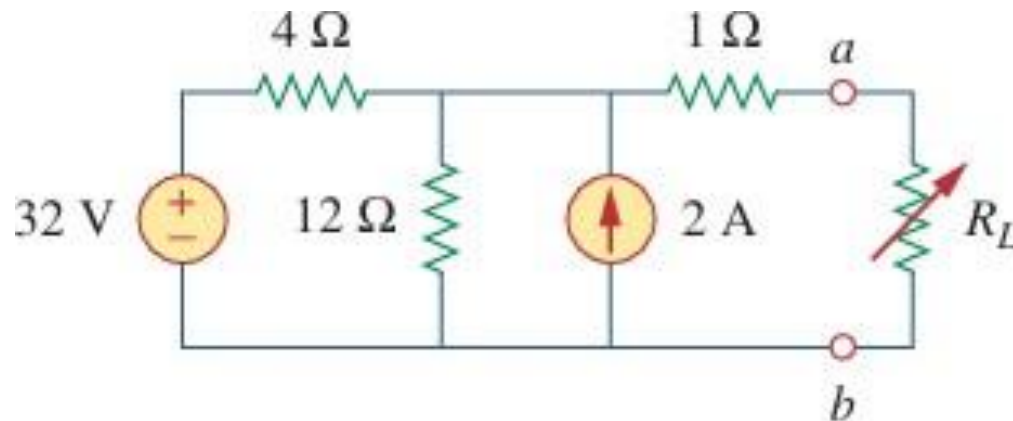
$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$V_L = R_L I_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$

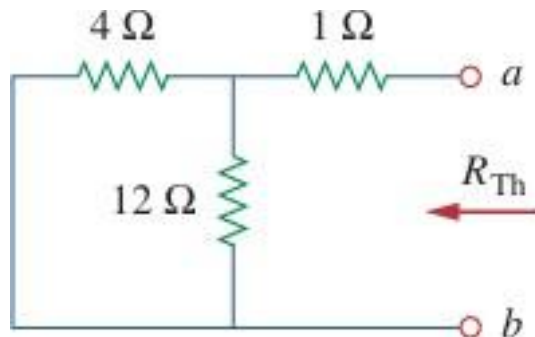
Thevenin's Theorem

Find the Thevenin's equivalent circuit of the circuit shown below, to the left of the terminals.

Then find the current through $R_L = 6\ \Omega$.



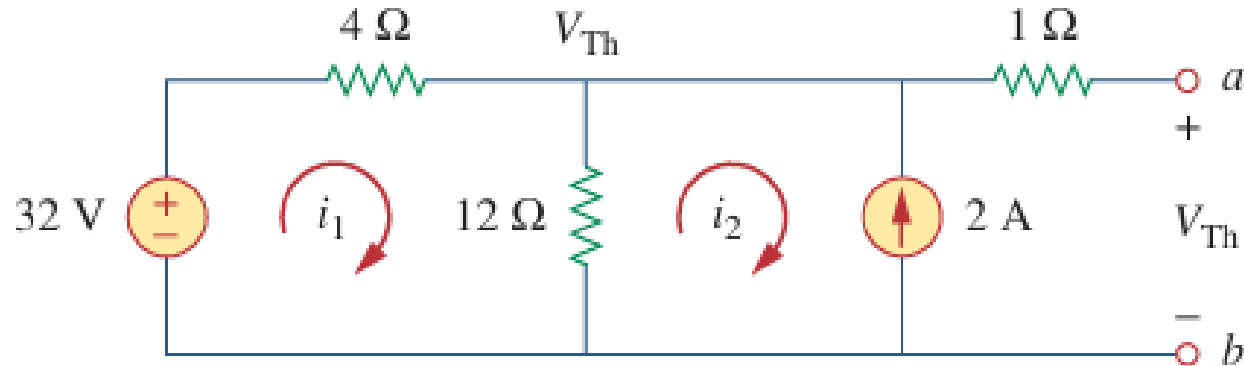
Find R_{th} by removing all the independent sources.



$$R_{Th} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = 4\ \Omega$$

Thevenin's Theorem

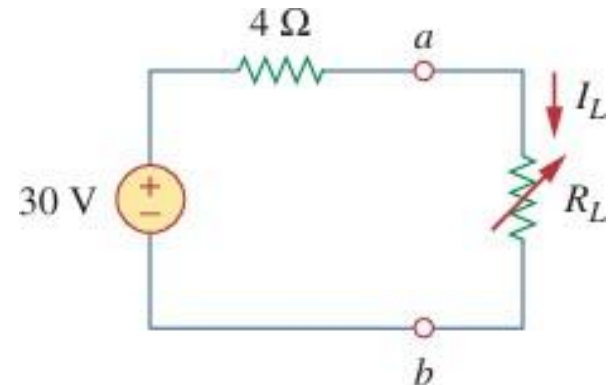
Find V_{th} by removing the load and finding voltage at the terminals a and b.



$$-32 + 4i_1 + 12(i_1 - i_2) = 0, \quad i_2 = -2 \text{ A}$$

$$i_1 = 0.5 \text{ A.}$$

$$V_{Th} = 12(i_1 - i_2) = 12(0.5 + 2.0) = 30 \text{ V}$$



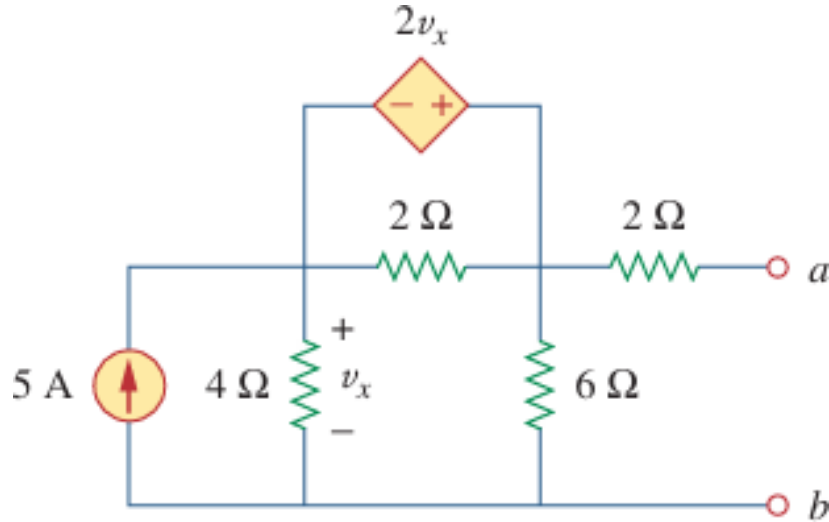
When $R_L = 6$,

$$I_L = \frac{30}{10} = 3 \text{ A}$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

Thevenin's Theorem

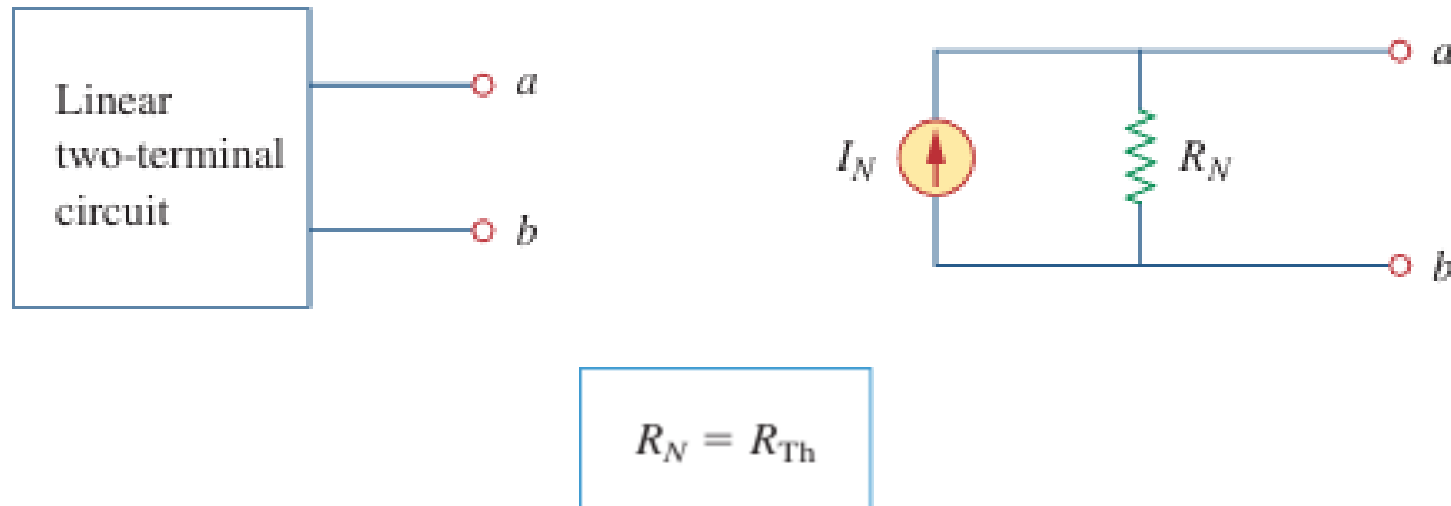
Find the Thevenin's equivalent of the circuit at terminals a-b.



1. To find R_{th} we set the independent source equal to zero but leave the dependent source alone.
2. However, we excite the network with a voltage source v_o connected to the terminals.
3. For easy calculation set $v_o = 1$ V.
4. Let us now find the current i_o through the terminals, and then obtain $R_{th} = v_o/i_o$

Norton's Theorem

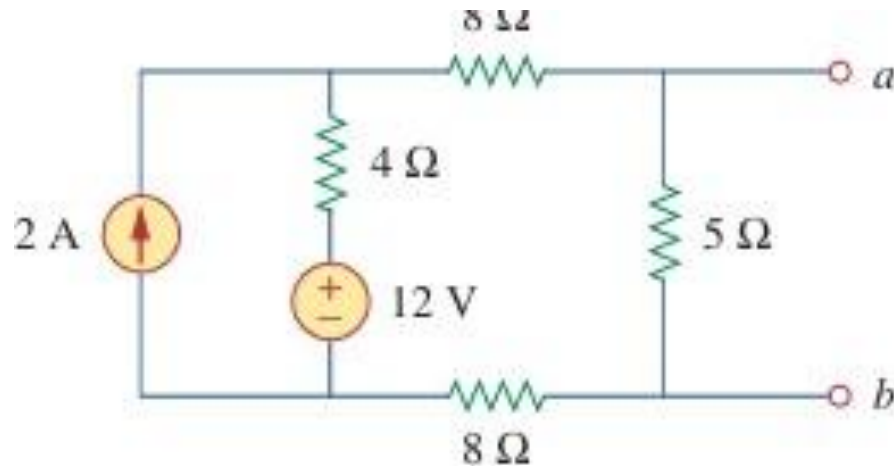
Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.



Two circuits are said to be equivalent if they have the same voltage-current relation at their terminals.

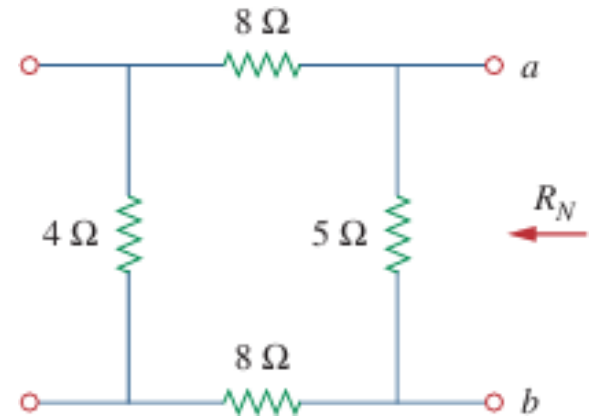
Norton's Theorem

Find the Norton equivalent circuit of the circuit below at terminals a-b.



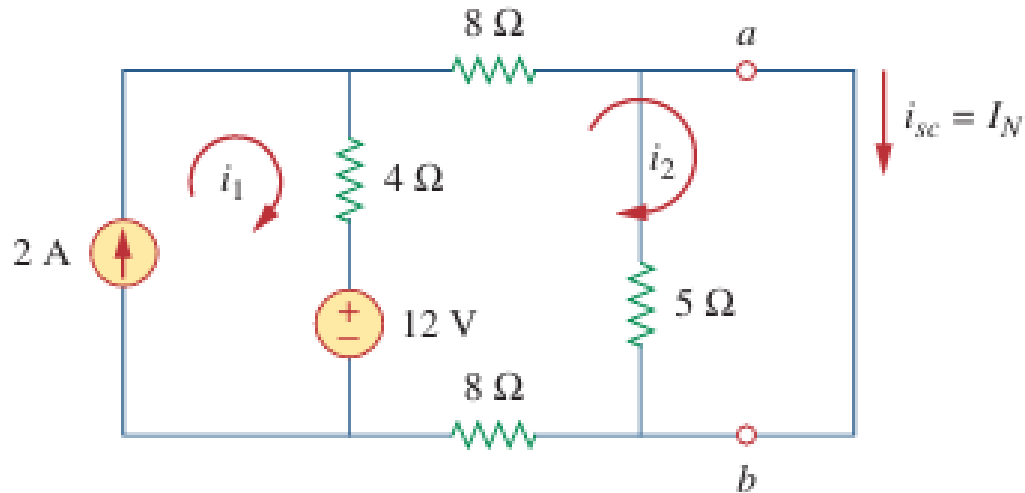
R_N

$$R_N = 5 \parallel (8 + 4 + 8) = 5 \parallel 20 = \frac{20 \times 5}{25} = 4 \Omega$$



Norton's Theorem

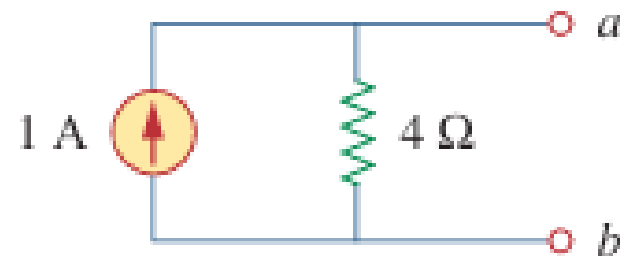
I_N



$$i_1 = 2 \text{ A}$$

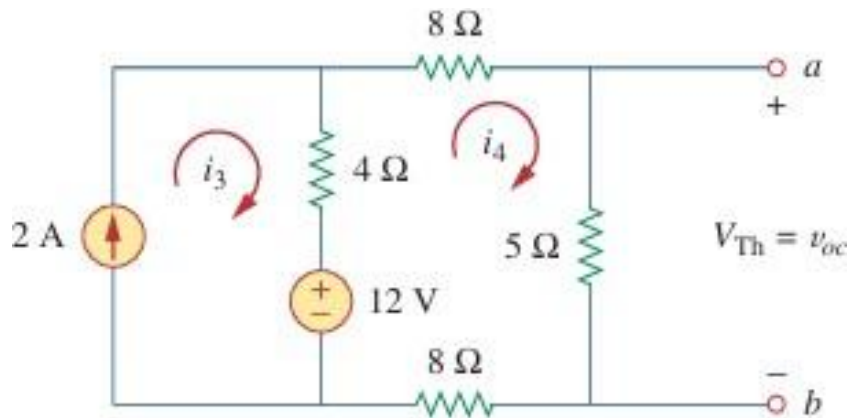
$$20i_2 - 4i_1 - 12 = 0$$

$$i_2 = 1 \text{ A} = i_{sc} = I_N$$



Norton's Theorem

Alternatively, we can also determine I_N from V_{th} .



$$I_N = \frac{V_{Th}}{R_{Th}}$$

$$i_3 = 2 \text{ A}$$

$$25i_4 - 4i_3 - 12 = 0 \quad \Rightarrow \quad i_4 = 0.8 \text{ A}$$

$$v_{oc} = V_{Th} = 5i_4 = 4 \text{ V}$$

$$I_N = \frac{V_{Th}}{R_{Th}} = \frac{4}{4} = 1 \text{ A}$$